

Emil Kim

Busan, South Korea, 46286

emilkim01@pusan.ac.kr • linkedin.com/in/emeelkaa • emeelkaa.github.io • +82-10-2892-7911

Education

Pusan National University

M.Sc. student in Artificial Intelligence
Graduate School Scholarship

Busan, South Korea
Mar 2025 – Present

Pusan National University

B.Sc. in Computer Science and Engineering
Samsung Dream Scholarship (2 recipients selected per university) • Outstanding Graduate

Busan, South Korea
Sep 2020 – Jan 2025

Research Interests

Biosignal Analysis (EEG, ECG) • Multimodal Representation • Medical Image Analysis.

Research Experience

IC&ML Lab, Pusan National University

Graduate Research Assistant

Busan, South Korea
Sep 2024 – Present

- Developed multimodal learning frameworks for EEG–fMRI translation, focusing on cross-modal representation learning, achieving superior reconstruction performance and reducing model parameters by $3.2\times$ and peak memory by $2.3\times$.
- Designed, trained and evaluated deep learning architectures based on Transformers and state-space models for neural signal analysis and representation learning, improving performance by 6.6% and reducing inference latency by 7.1% across benchmark datasets.
- Developed and evaluated a CNN-based model for Parkinson’s disease classification from SPECT images, incorporating saliency map visualization to interpret model predictions.

Teaching Experience

Pusan National University

Teaching Assistant — Introduction to Unix Programming (CB2001102-060)

Busan, South Korea
Mar 2026 – Jun 2026

- Designed programming assignments, led weekly practice sessions, and mentored undergraduate students.

Publications

Journal Articles

- E. Kim and J. K. Gahm, “HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding,” *IEEE Access*, vol. 14, pp. 79188–79196, 2026.
- E. Kim, Y. Kenes, S. D. Tran, and J. K. Gahm, “Multi-Scale EEG Feature Extraction with Attention Mechanism for Seizure Detection,” *Journal of KIISE*, vol. 53, no. 6, pp. 497–506, 2026.

Conference Proceedings

- E.E. Kim and J. K. Gahm, “MARBLE: Lightweight EEG-to-fMRI Translation via Mamba-Attention and ROI-Conditioned Decoding,” *MICCAI*, Strasbourg, France, Sep 2026.
- E. Kim and J. K. Gahm, “CGM-EEG: Cross-Gated Mamba for Spatio-Temporal EEG Representation Learning,” *ISBI (Oral)*, London, UK, Apr 2026.
- E. Kim, Y. Kenes, S. D. Tran, and J. K. Gahm, “Multi-Scale EEG Feature Extraction with Attention Mechanism for Seizure Detection,” *Korea Computer Congress*, Jeju, South Korea, Jul 2025.

Awards & Scholarships

1st Place, Three Minute Thesis (3MT), Pusan National University, South Korea
Best Paper Award, Physical AI International Forum (JKAIA), Seoul, South Korea
Graduate School Scholarship, Pusan National University, Graduate School of AI
Samsung Dream Scholarship, Samsung Dream Scholarship Foundation

Aug 2026
Nov 2025
Mar 2025 – Present
Sep 2024 – Dec 2025

Skills & Interests

Programming: Python, C, C++, MATLAB

ML Frameworks: PyTorch, Hugging Face, Scikit-Learn

Tools: Git, Linux, Docker, CUDA, NumPy, Pandas

Language: Russian (Native), English (IELTS 8.0), Korean (TOPIK Level 5)